Yewon Jang

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EDUCATION

Hanyang University Seoul, South Korea
B.S. in Mechanical Engineering Mar 2022 – Aug 2026
Relevant Courses: Strength of Materials, CAD, Dynamics, Engineering Mathematics, Numerical Analysis, Fluid Mechanics,

Ohio Northern University Ada, OH, USA
Exchange Program in Mechanical Engineering, Aug 2024 – May 2025
Relevant Courses: Finite Element Analysis

RESEARCH INTERESTS

Biomechanics: Extending patient-specific 3D modeling from linear-elastic toward hyper-elastic and contact-mechanics frameworks to capture complex material and interface behaviors more accurately.
Computational Mechanics: Developing physics-informed machine learning and surrogate models to overcome the computational costs and convergence bottlenecks of traditional finite element solvers.

PUBLICATIONS AND PRESENTATIONS

Yewon Jang, Colleen H. Kottmyer, Hui Shen, "A Finite Element Analysis Comparing the Biomechanical Stability of Fascia-Wrapped versus PRF-Embedded Diced Cartilage Grafts in Dorsal Augmentation Rhinoplasty", Proceedings of ASEE – NCS Conference, 2026, Alliance, OH

Cheng, X., Geibel, H. G., Clawson, S. J., **Jang, Y**., Hastings, T. C., Guerra, F. S., Bak, P. D., Yang, G., Herdman, V. T., Betts, C. M., and Shen, H., 2025, "A Photogrammetry-Based Approach for Patient-Specific Modeling in Syndactyly Surgery: Prototyping, 3D Reconstruction, and Finite Element Analysis," Proceedings of the ASME 2025 International Mechanical Engineering Congress and Exposition (IMECE2025), Memphis, TN, USA, Nov. 16–20, Paper No. IMECE2025-166019.

Yang, G., Shen, H., **Jang, Y**., and Cheng, X., "Finite Element Analysis-Assisted Surgical Planning and Evaluation of Flap Design in Hand Surgery," Frontiers in Bioengineering and Biotechnology, 13 (2025): 1611993

RESEARCH EXPERIENCE

Hanyang University & Ohio Northern University Seoul (Remote) & OH (In-person)
Undergraduate Research Assistant /Mechanical Engineering Aug 2024 – Jan 2026

Project: Biomechanical Analysis of Rhinoplasty Graft Stability
Compared the biomechanical responses of DC-F and DC-PRF grafts in rhinoplasty using FEA, focusing on stress and deformation under loading conditions.

- **Lead Researcher:** led the end-to-end research workflow, from study design and 3D modeling to manuscript preparation.
- **Advanced Modeling:** Developed 3D nasal models in ANSYS, applying the Rule of Mixtures to calculate effective elastic moduli for composite grafts.
- **Postoperative Stage Modeling:** Modeled changes in graft stiffness and interface conditions at three postoperative stages: immediately after surgery, one month, and one year.
- **Key Findings:** Found the DC-F exhibited approximately 10–38 times higher maximum von Mises stress than DC-PRF at one month, depending on leading conditions and cartilage fraction, while DC-PRF showed greater sensitivity to changes in applied load.

Project: Syndactyly Surgery Optimization via Photogrammetry & FEA
Contributed FEA modeling to a collaborative project combining a photogrammetry-based hand reconstruction with surgical flap analysis

- **Simulation Lead:** Performed structural analysis in ANSYS, including meshing and boundary-condition setup from patient anatomical data on hand models reconstructed.
- **Flap Analysis:** Analyzed hexagonal-flap biomechanics during suturing, extracting patient-specific flap dimensions (h, w) using anatomy derived dimensions.
- **Convergence Solution:** Addressed 3D contact and convergence challenges by applying a 3D-to-2D surface-unrolling method to ensure numerical stability.
- **Reliability Engineering:** Conducted Mesh Convergence Studies to validate the numerical stability of surgical models.

Project: Review of FEA Applications in Hand Surgery
Reviewed FEA-assisted surgical planning for hand skin flaps, highlighting current trends in patient-specific modeling and AR/MR integration.

- **Literature Synthesis:** Compiled material models, geometries, and boundary/loading conditions from FEA flap-design studies (2000–2025) into comparative tables.
- **Gap Identification:** Identified skin-property characterization and cross-study validation as the field's central limitations, informing the review's recommendation for AI-driven, adaptive flap-design tools.

ACADEMIC PROJECTS

Undergraduate Thesis: Multi-Linked Wheeled Snake Robot Spring 2025
Developed an 8-segment hybrid robot and a Python-based spatial path-following algorithm for high-precision trajectory tracking.

Braille Display Device for Korean Text Conversion Winter 2026
Designed a solenoid-actuated mechatronics system to convert real-time Korean text input into tactile Braille patterns.

Internal Combustion Engine Performance Analysis Spring 2024
Analyzed engine efficiency (IMEP, ISFC) and thermal performance based on Start of Injection (SOI) variables using P-V diagrams.

SCHOLARSHIPS

Scholarship	Institution	Year	Semester
Merit Based Scholarship	Hanyang University	2023	Spring Semester
National Scholarship (Co-op) Of Creative Korea 2	Hanyang University	2023	Spring Semester
Korea-U.S. Student Exchange Program in the field of high-tech industry	Korea Institute for Advancement of Technology (KIAT)	2024-2025	Fall Semester – Spring Semester

SKILLS

Software & Programming: ANSYS Mechanical, ANSYS SpaceClaim, ANSYS DesignModeler, CATIA, SolidWorks, Python, MATLAB, Microsoft Excel

TEST SCORES

GRE General Test: Q: 170, V: 151, AW: 3.5
TOEFL iBT: 5.0/6.0 (Reading: 4.5, Listening: 5.0, Speaking: 4.5, Writing: 5.5) | *Oct 2026*